

EEE4114F: Digital Signal Processing

Class Test

11 March 2020

SOLUTIONS

Name:

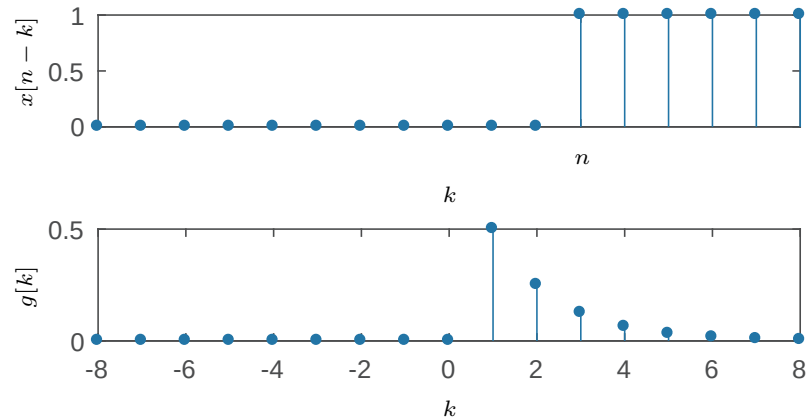
Student number:

Information

- The test is closed-book.
 - This test has *four* questions, totalling 20 marks.
 - Answer *all* the questions.
 - You have 60 minutes.
 - An information sheet is attached.
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1. (5 marks) Suppose $x[n] = u[-n]$ and $g[n] = (\frac{1}{2})^n u[n-1]$. Find and plot $y[n] = g[n] * x[n]$.

The required result is most easily found using graphical methods applied to the relation $y[n] = \sum_{k=-\infty}^{\infty} g[k]x[n-k]$, with each component signal represented below:



For $n \leq 1$ we have $y[n] = \frac{1}{2} + \frac{1}{4} + \dots = 1$. For $n > 1$ the output is

$$\begin{aligned} y[n] &= \sum_{k=n}^{\infty} (1/2)^k = (1/2)^n + (1/2)^{n+1} + (1/2)^{n+2} + \dots \\ &= (1/2)^n (1 + 1/2 + 1/4 + \dots) = 2(1/2)^n. \end{aligned}$$

2. (5 marks) Consider a causal filter defined by the following constant coefficient difference equation:

$$y[n] - 1.5y[n-1] + 0.5y[n-2] = x[n].$$

- (a) Suppose the input is $x[n] = \delta[n]$ and the system is initially at rest. Find and plot the output $y[n]$ over the range $-5 \leq n \leq 5$.
- (b) Is the system stable? Why?
- (c) What is the DC gain of the system?

- (a) The input is zero for $n < 0$. Since the system is causal the initial rest condition means that $y[-1] = y[-2] = 0$. Using

$$y[n-2] = 2(x[n] - y[n] + 1.5y[n-1])$$

it is then easy to see that $y[n] = 0$ for all $n < 0$. For $n \geq 0$ we can simply find the output from the forward recursion $y[n] = 1.5y[n-1] - 0.5y[n-2] + x[n]$ and plot the result:

$$\begin{aligned} y[0] &= x[0] = 1 \\ y[1] &= 1.5(1) = 1.5 \\ y[2] &= 1.5(1.5) - 0.5(1) = 1.75 \\ y[3] &= 1.5(1.75) - 0.5(1.5) = 1.875 \\ y[4] &= 1.5(1.875) - 0.5(1.75) = 1.9375 \\ y[5] &= 1.5(1.9375) - 0.5(1.875) = 1.9688. \end{aligned}$$

- (b) The z-transform of the LCCDE gives

$$Y(z) - 1.5z^{-1}Y(z) + 0.5z^{-2}Y(z) = Y(z)(1 - 1.5z^{-1} + 0.5z^{-2}) = X(z),$$

so the system function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{(1 - z^{-1})(1 - 0.5z^{-1})} = \frac{z^2}{(z - 1)(z - 0.5)}$$

Since one of the poles is on the unit circle the system is not stable.

- (c) The system function does not converge on the unit circle. It therefore does not have a frequency response and the DC gain is not defined.

3. (5 marks) Answer the following:

- (a) Find the DTFT of the signal $x[n] = (\frac{1}{4})^n u[n+2]$.
- (b) Suppose a filter has an impulse response $h[n] = (\frac{1}{4})^n u[n+2]$. Is the filter stable? Why?
- (c) Let $X(e^{j\omega})$ denote the DTFT of a complex sequence $x[n]$. Determine the DTFT $Y(e^{j\omega})$ of the sequence $y[n] = x[n] * x^*[-n]$ in terms of $X(e^{j\omega})$, and show that it is a real-valued function of ω .

- (a) Noting that $x[n] = (1/4)^{-2}(1/4)^{n+2}u[n+2] = 16(1/4)^{n+2}u[n+2]$ we can use the time shift property to find

$$X(e^{j\omega}) = 16e^{j2\omega} \frac{1}{1 - \frac{1}{4}e^{-j\omega}}$$

- (b) The filter is stable if the impulse response is absolutely summable. Now

$$\begin{aligned} S &= \sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=-\infty}^{\infty} (1/4)^n u[n+2] = \sum_{n=-2}^{\infty} (1/4)^n = \sum_{k=0}^{\infty} (1/4)^{k-2} \\ &= 16 \sum_{k=0}^{\infty} (1/4)^k = \frac{16}{1 - 1/4} < \infty, \end{aligned}$$

so the system is stable.

- (c) Suppose $v[n] = x^*[-n]$. Then

$$\begin{aligned} V(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x^*[-n]e^{-j\omega n} = \left(\sum_{n=-\infty}^{\infty} x[-n]e^{j\omega n} \right)^* \\ &= \left(\sum_{m=-\infty}^{\infty} x[m]e^{-j\omega m} \right)^* = X^*(e^{j\omega}). \end{aligned}$$

Therefore

$$Y(e^{j\omega}) = X(e^{j\omega})X^*(e^{j\omega}) = |X(e^{j\omega})|^2,$$

which is a real-valued quantity.

4. (5 marks) For an LTI system we are given the Z-transforms of the input and output signals

$$X(z) = \frac{1}{1 + z^{-2} + z^{-4}} \quad \text{and} \quad Y(z) = \frac{1}{2} + \frac{1}{4}z^{-1}.$$

- (a) Determine and sketch the impulse response $h[n]$
- (b) Determine the DTFT of $h[n]$
- (c) Determine and sketch a pole-zero plot of the system and ROC in the z-plane.

- (a) The system function is

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} = \left(\frac{1}{2} + \frac{1}{4}z^{-1} \right) (1 + z^{-2} + z^{-4}) \\ &= \frac{1}{2} + \frac{1}{4}z^{-1} + \frac{1}{2}z^{-2} + \frac{1}{4}z^{-3} + \frac{1}{2}z^{-4} + \frac{1}{4}z^{-5}. \end{aligned}$$

The impulse response is the inverse z-transform

$$h[n] = \frac{1}{2}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n-2] + \frac{1}{4}\delta[n-3] + \frac{1}{2}\delta[n-4] + \frac{1}{4}\delta[n-5].$$

- (b) The system function $H(z)$ has no poles (except perhaps at the origin) so the ROC is the entire z-plane. The z-transform evaluated on the unit circle gives the DTFT:

$$H(e^{j\omega}) = \frac{1}{2} + \frac{1}{4}e^{-j\omega} + \frac{1}{2}e^{-j2\omega} + \frac{1}{4}e^{-j3\omega} + \frac{1}{2}e^{-j4\omega} + \frac{1}{4}e^{-j5\omega}.$$

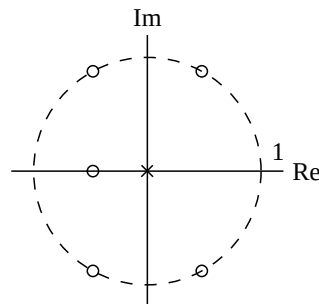
- (c) The z-transform can be written as follows:

$$H(z) = \frac{1}{2z^5} (z + 1/2) (z^4 + z^2 + 1) = \frac{1}{2z^5} (z + 1/2) (z^2 - a_s)(z^2 - a_s^*)$$

with $a_s = -1/2 + j\sqrt{3}/2 = e^{j2\pi/3}$. Thus

$$H(z) = \frac{1}{2z^5} (z + 1/2) (z - e^{j\pi/3})(z + e^{j\pi/3})(z - e^{-j\pi/3})(z + e^{-j\pi/3})$$

and the pole-zero plot follows:



Transforms

$$\begin{aligned}
 X(j\Omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt & \iff & x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega & \text{(CTFT)} \\
 X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} & \iff & x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega & \text{(DTFT)} \\
 X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} & \iff & x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz & \text{(ZT)} \\
 X[k] &= \sum_{n=0}^{N-1} x[n]W_N^{kn} & \iff & x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} & \text{(DFT)}
 \end{aligned}$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n + 1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$